Emerson University Multan

Computer Department

Bs. Software Engineering (Morning)

Group : 15

Subject :

Web Technologies

Project Topic :

Islamic Quiz App

Submitted To :

Dr. Zahid Aziz

Submitted By :

Aawaiz Tayyab (05)

AbduRehman (52)

M Saad (31)

Date:_____

Table Of Content

1-Introduction

2-Purpose of the Project

3-Tools and Technologies Used

4-System Design

5-Implementation

6-Question Bank Structure

7-Working of the Application

8-Use Case Diagram

9-Conclusion

1- Introduction

The **Islamic Quiz Application** is a **web-based educational platform** designed to help users learn and test their knowledge about Islam in an **interactive and engaging manner**. Traditional learning methods, such as reading books or attending lectures, often do not provide immediate feedback or interactive engagement. This application bridges that gap by combining **Islamic education with modern web technologies**, making learning both **fun and effective**.
The quiz is divided into three main categories:

- **Pillars of Islam** – covering the five essential acts of worship and their significance.
- **Prophets** – focusing on key prophets in Islamic history and their teachings.
- **Quran** – testing knowledge about the holy scripture, including its chapters and messages.

Users can select a category, answer multiple-choice questions, and receive **instant feedback**. Correct answers are highlighted in green, and incorrect answers are shown in red, allowing users to **learn from their mistakes immediately**. At the end of the quiz, the application calculates the user's score and displays it as a percentage, motivating users to improve their performance through repeated attempts.
The application also demonstrates practical **web development skills** using :

- **HTML** – for structure.
- **CSS** – for styling.
- **Js** – for dynamic functionality.

The interface is designed to be **user-friendly and responsive**, ensuring accessibility on desktops, tablets, and mobile devices.
In conclusion, the Islamic Quiz Application is more than a quiz; it is a **learning platform** that combines **education, interactivity, and technology** to enhance users' understanding of Islam. It is suitable for students, teachers, and anyone interested in Islamic studies, providing a **fun, engaging, and effective way to learn**.

2- Purpose of the Project

The primary purpose of the **Islamic Quiz Application** is to serve as an **educational tool** that allows users to **learn and evaluate their understanding of Islam** in an interactive and engaging way. Unlike traditional learning methods, such as reading books or listening to lectures, this application encourages **active participation** by presenting information in the form of quizzes. This interactive approach helps users retain knowledge better and makes the learning process more enjoyable.
The application focuses on three key areas of Islamic knowledge:

1. **Pillars of Islam** – This category covers the **five fundamental acts of worship** that form the foundation of a Muslim's faith: Shahada (faith declaration), Salah (prayer), Zakat (charity), Sawm (fasting), and Hajj (pilgrimage). Through questions related to these pillars, users can deepen their understanding of their significance, rules, and practical application in daily life.
2. **Prophets** – This category focuses on the **major prophets in Islamic history**, their teachings, and notable events from their lives. Users can learn about the stories of prophets like Muhammad (PBUH), Musa, Ibrahim, and others, understanding their moral lessons and the guidance they provided to their communities.
3. **Quran** – This category tests knowledge about the **holy scripture of Islam**, including the number of chapters (Surahs), verses, themes, and key teachings. Users can also learn about the historical context of revelation, the message of different Surahs, and important lessons from the Quran.

By using this application, users are **actively engaging with content** rather than passively reading it. The quiz format promotes **critical thinking, memory retention, and self-assessment**, allowing users to identify areas where they need to improve and encouraging repeated practice. Additionally, the instant feedback provided for each answer reinforces correct knowledge and clarifies misconceptions immediately.
In essence, the purpose of the Islamic Quiz Application is not only to **test knowledge** but also to **educate, engage, and motivate learners**. It combines **Islamic teachings with modern technology**, providing a **structured, interactive, and user-friendly learning experience** for students, teachers, and anyone interested in expanding their understanding of Islam.

3- Tools and Technologies Used

Tool/Technology	Purpose
HTML	Structure and content of the web pages
CSS	Styling, layout, and responsive design
JavaScript	Quiz logic, interactivity, and score calculation
Browser (Chrome/Edge)	Run and test the application
Text Editor (VS Code)	Code writing and management

The Islamic Quiz Application is built using the following tools and technologies:

- **HTML5** – Provides the **structure and content** of the application, including the Welcome, Quiz, and Result screens. Key elements like headings (`<h1>`, `<h2>`), paragraphs (`<p>`), buttons (`<button>`), and containers (`<div>`) are created with HTML
- **CSS3** – Styles the application, making it **visually appealing** and **responsive** on different devices. In this project, CSS is responsible for Styling **quiz cards, buttons, and containers** to make the interface appealing.Highlighting **correct and incorrect answers** using colors like green and red.Making the application **responsive** so it works on desktops, tablets, and mobile devices.Essentially, CSS ensures that the application is **not only functional but also user-friendly and visually attractive**.
- **JavaScript (ES6)** – Adds **interactivity**, including loading questions, checking answers, and calculating scores.
- **Browser (Chrome/Edge)** – Used to **run and test** the application to ensure it works correctly.
- **Text Editor (VS Code)** – Used for **writing and managing code** efficiently.

Summary:

These technologies work together to create a **functional, interactive, and user-friendly quiz application**.

4- System Design

4.1 Architecture

The system uses **client-side architecture**, meaning all logic runs in the user's browser.

Flow of Application:

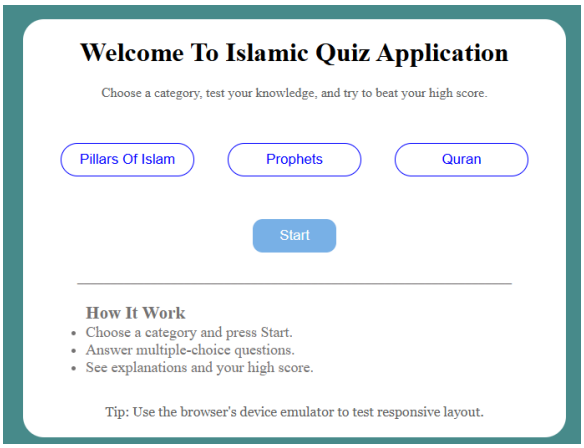
User → Select Category → Start Quiz → Load Questions → Select Answer → Next Question → End Quiz → Result Screen → Retry/Home

4.2 User Interface Design

The UI consists of three main screens:

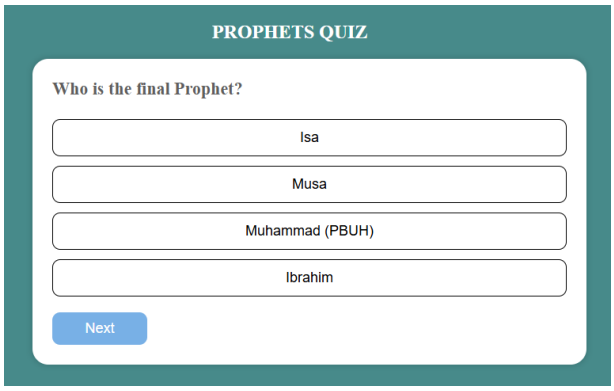
4.2.1 Welcome Screen:

- Introduction to the quiz.
- Category selection buttons.
- Start button.



4.2.2 Quiz Screen:

- Question displayed at the top.
- Four options for answers.
- Next button to proceed.



4.2.3 Result Screen:

- Shows total score and percentage.
- Retry and Home buttons.

5- Implementation

5.1-HTML Structure

- HTML uses `<div>` for containers, `<button>` for options, and `<p>` for text.
- Three main screens: **Welcome**, **Quiz**, and **Result**.
- IDs and classes are used for styling and JavaScript functionality.

Example HTML Snippet:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <link rel="stylesheet" href="main.css">
</head>
<body>
    <div class="mainbody">
        <h1>Welcome To Islamic Quiz Application</h1>
        <p>Choose a category, test your knowledge, and try to beat your high score.</p>
        <div class="btns">
            <button class="btn" id="b1">Pillars Of Islam</button>
            <button class="btn" id="b2">Prophets</button>
            <button class="btn" id="b3">Quran</button>
        </div>/.btns
        <button class="start_btn">Start</button>
        <p>_____</p>
        <div class="text">
            <div class="text1">
                <h3>How It Work</h3>
                <ul>
                    <li>Choose a category and press Start.</li>
                    <li>Answer multiple-choice questions.</li>
                    <li>See explanations and your high score.</li>
                </ul>
            </div>/.text1

        </div>/.text
        <div class="tip">Tip: Use the browser's device emulator to test responsive layout.</div>
    </div>/.mainbody
```

5.2-CSS Styling

- `.mainbody` for main container.
- `.btn` for category buttons with hover effects.
- `.optionBtn` for answer options.
- `.correct` and `.wrong` classes highlight answers.

Example CSS Snippet:

```css
*{
    margin: 0px;
    padding: 0px;
    border: border-box;
}
body{
    background-color: rgb(71, 138, 138);
    display: flex;
    justify-content: center;
}
.mainbody{
    height: 500px;
    width: 650px;
    background-color: white;
    border-radius: 25px;
    display: flex;
    justify-content: space-evenly;
    flex-direction: column;
    align-items: center;
    margin-top: 70px;
}
h1{
    margin: 0;
}
.btns{
    height: 70px;
    width: 640px;
    display: flex;
    justify-content: space-evenly;
    margin-top: 30px;
}
.btn{
```

5.3-JavaScript Functionality

- Variables store category, question index, score, and question bank.
- Functions include:

1. `selectCategory()` – Highlights selected category.
2. `startBtn.onclick` – Starts quiz and loads questions.
3. `loadQuestion()` – Loads current question and options.
4. `checkAnswer()` – Checks user answer and updates score.
5. `nextBtn.onclick` – Loads next question or ends quiz.
6. `endQuiz()` – Displays final score and percentage.
7. `retryBtn.onclick` – Restarts quiz.
8. `homeBtn.onclick` – Returns to welcome screen.

Example JavaScript Snippet:

Welcome | main.html | main.css | script.js

script.js > ...

```javascript
// SELECTING HTML ELEMENTS
const b1 = document.getElementById("b1");
const b2 = document.getElementById("b2");
const b3 = document.getElementById("b3");
const startBtn = document.querySelector(".start_btn");

const welcomeScreen = document.querySelector(".mainbody");
const quizScreen = document.getElementById("quizScreen");
const resultScreen = document.getElementById("resultScreen");

let selectedCategory = "";
let score = 0;
let qIndex = 0;
let currentQuestions = [];

// QUIZ ELEMENTS
const categoryTitle = document.getElementById("categoryTitle");
const questionText = document.getElementById("questionText");

const opt1 = document.getElementById("opt1");
const opt2 = document.getElementById("opt2");
const opt3 = document.getElementById("opt3");
const opt4 = document.getElementById("opt4");

const nextBtn = document.getElementById("nextBtn");

// RESULT ELEMENTS
const finalScore = document.getElementById("finalScore");
const retryBtn = document.getElementById("retryBtn");
const homeBtn = document.getElementById("homeBtn");

```

6- Question Bank Structure

- Stored as a **JavaScript object**.
- Each category has **5 questions**, extendable to more.
- **Structure:**

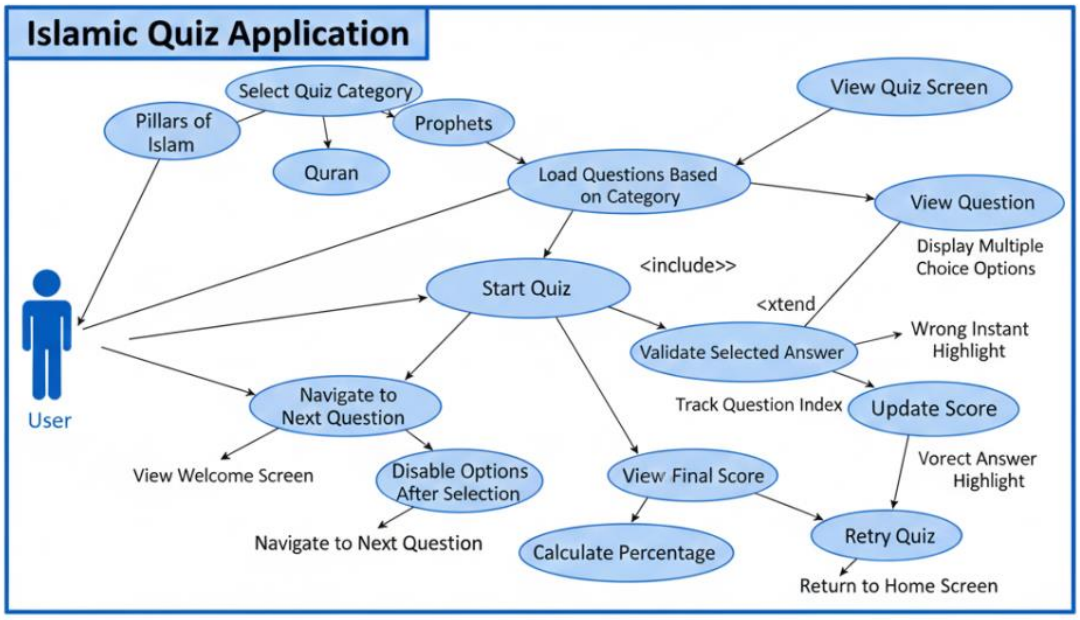
```
Welcome   main.html   main.css   script.js
script.js > ...

// QUESTION BANK (5 QUESTIONS EACH)
const questionBank = {
    pillars:[
        {q:"How many pillars are there in Islam?", a:["4","5","6","7"], c:1},
        {q:"What is the first pillar?", a:["Salah","Shahada","Zakat","Hajj"], c:1},
        {q:"Zakat means?", a:["Charity","Prayer","Fasting","Pilgrimage"], c:0},
        {q:"Which month is fasting?", a:["Shawwal","Rajab","Ramadan","Muharram"], c:2},
        {q:"Hajj is performed in?", a:["Madina","Makkah","Taif","Yemen"], c:1},
        {q:"Which pillar requires fasting during Ramadan?", a:["Salah","Zakat","Sawm","Hajj"], c:2},
        {q:"Shahada is the declaration of...", a:["Charity","Faith","Prayer","Pilgrimage"], c:1},
        {q:"How many times a day should Muslims pray?", a:["3","5","7","2"], c:1},
        {q:"Zakat is obligatory for whom?", a:["All Muslims","Only Poor","Only Rich","Only Adults"], c:2},
        {q:"Which pillar involves pilgrimage to Makkah?", a:["Salah","Zakat","Hajj","Sawm"], c:2},

    ],
    prophets:[
        {q:"Who is the final Prophet?", a:["Isa","Musa","Muhammad (PBUH)","Ibrahim"], c:2},
        {q:"Who built an Ark?", a:["Adam","Nuh","Yunus","Idris"], c:1},
        {q:"Who split the sea?", a:["Isa","Musa","Yusuf","Ibrahim"], c:1},
        {q:"Who was swallowed by a fish?", a:["Musa","Yunus","Zakariya","Nuh"], c:1},
        {q:"Who was known for patience?", a:["Ayub","Idris","Saleh","Yusuf"], c:0},
        {q:"Who is the father of Prophet Ismail?", a:["Ibrahim","Adam","Musa","Nuh"], c:0},
        {q:"Who was known for wisdom and judgment?", a:["Sulaiman","Yusuf","Musa","Muhammad"], c:0},
        {q:"Which Prophet was thrown into the fire but remained safe?", a:["Ibrahim","Musa","Isa","Nuh"],
        {q:"Who received the Ten Commandments?", a:["Isa","Musa","Yusuf","Ibrahim"], c:1},
        {q:"Which Prophet interpreted dreams in Egypt?", a:["Yusuf","Musa","Ibrahim","Zakariya"], c:0},

    ],
    quran:[
        {q:"How many Surahs in Quran?", a:["114","200","90","50"], c:0},
        {q:"The Quran was revealed in?", a:["Arabic","Persian","Hebrew","Turkish"], c:0},
```

- Each question contains:
  - `q` → question text
  - `a` → array of 4 options
  - `c` → index of correct answer

7- Working of the Application

1. User selects a category.
2. Clicks **Start** → first question loads.
3. User selects an answer → feedback shown.
4. Click **Next** → next question or end of quiz.
5. At the end → Result screen shows **score and percentage**.
6. Options to **Retry** or return to **Home**.

8- Use Case Diagram



9- Conclusion

The Islamic Quiz Application successfully demonstrates a **web-based educational tool**.

- Users can learn and test their knowledge of Islam interactively.
- Provides **immediate feedback** on answers.
- Implements **HTML, CSS, and JavaScript** effectively.
- Can be extended with **timer, high score tracking, or explanations**.

This project enhances both **programming skills** and **UI/UX design** experience.

Thank You